Wyatt Walsh

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WORK EXPERIENCE

Associate III, Senior Software Engineer
May 2023 – Present • JPMorgan Chase & Co.

Designed high-performance, scalable data integrations for Credit Technology within Commercial & Investment Banking (CIB), ensuring timely delivery of critical market intelligence to traders and risk teams.

- Accelerated SDLC cycles and reduced debugging time across credit technology initiatives by using AI-driven development tools including Copilot, Codeium, and GPT-based LLMs.
- Engineered enterprise-grade, fast OOP architectures incorporating rigorous testing and best-in-class SDLC practices for mission-critical financial systems.
- Implemented machine learning models to identify financial risks and operational inefficiencies, including an ML-driven framework for detecting anomalies in data ingestion volumes.
- Architected scalable AWS cloud solutions and optimized performance across diverse database systems (SQL, NoSQL, Timeseries) essential for financial data processing.
- Streamlined data access for traders and analysts by creating a simplified Python interface for KDB+, featuring access group tracking and convenience functions.

Senior Software Engineer
Mar 2022 – Jul 2022 • Stealth Web3 Startup

Oversaw front-end engineering for a DeFi NFT DAO platform supporting NFT-collateralized cryptocurrency loans, facilitating a fast interface for intricate blockchain transactions.

- Improved DeFi transaction throughput and user experience by architecting React/Next.js applications integrating Redux, TypeScript, and SCSS.
- Enabled rapid feature iteration in a dynamic startup environment by adopting agile development methodologies using Jira and Notion.
- Enhanced code reliability and mitigated regressions by instituting a test-driven development (TDD) workflow using Cypress, Jest, and StorybookJS.
- Ensured secure and streamlined DeFi smart contract operations through the ethers.js for blockchain interactions.
- Accelerated technical growth and advanced project deliverables by mentoring two junior software developers.

Chief Technology Officer
Jun 2021 – Mar 2022 • SandLabs

Founded and directed the technical vision for a Web3 enterprise for decentralized data solutions, overseeing product development lifecycles and integrating conventional content ecosystems with blockchain-based architectures.

- Secured $75K in Web3 grant funding via Ocean Protocol to advance a large-scale Reddit data scraping initiative using Web3 technologies.
- Directed the design and a comprehensive Next.js (TypeScript) platform, featuring a landing page, technical blog, and integrated content portal.
- Elevated project visibility by forging strategic alliances with Duke University co-founders and prominent Web3 influencers.
- Oversaw complex technical execution including multi-million-post data scraping in Python, configuration of Solidity smart contracts, and React TypeScript solutions.

Demand Forecasting
Jan 2020 – May 2020 • Gap Inc.

Led data-driven forecasting initiatives, using advanced analytics to improve demand prediction accuracy by 5%-25% across multiple retail divisions, directly impacting inventory planning and supply chain operations.

- Improved demand forecasting accuracy by 5%-25% across organizational divisions by implementing sophisticated methodologies, optimizing inventory management and reducing operational inefficiencies.
- Drove strategic organizational change by translating complex statistical insights into actionable business strategies through compelling data-backed presentations to senior leadership.
- Engineered scalable forecasting solutions by analyzing over 5 years of time-series data using MySQL and Apache Spark distributed computing frameworks.

Course Staff – Data 8: Foundations of Data Science
Jan 2019 – Dec 2019 • University of California, Berkeley

Delivered expert instruction and operational support for Berkeley's flagship 'Data 8: Foundations of Data Science' course, translating complex statistical and programming concepts for over 1,600 undergraduates.

- Provided comprehensive instruction in statistical foundations, Python programming, and data analysis methodologies to over 1,600 undergraduate students across three semesters in Berkeley's largest data science course.
- Designed and delivered 75+ engaging lectures and laboratory sessions covering Python programming, statistical inference, and data visualization techniques using real-world datasets.
- Orchestrated critical course operations including systematic management of office hours, examination protocols, assessment frameworks, and assignment evaluation across sections.

EDUCATION

Bachelor of Science in Industrial Engineering and Operations Research — Aug 2014 – Dec 2021 • University of California, Berkeley College of Engineering

High School Diploma — Aug 2012 – May 2014 • The Hotchkiss School

PROJECTS

Personal Website: w4w.dev
Nov 2018 – Present

Personal portfolio & technical blog (V6) built with Next.js/TypeScript/TailwindCSS/shadcn-ui. Features advanced MDX content processing (30+ plugins), KaTeX math rendering, tsParticles animations, YOURLS integration (w4w.to), deployed on Vercel.

- Architected and iterated on a personal website platform through six major versions (450+ commits), migrating from Hugo to Next.js App Router and demonstrating continuous technical evolution.
- Engineered an extensive MDX content processing pipeline using 30+ remark/rehype plugins for advanced Markdown features, syntax highlighting, auto-generated TOCs, and custom components.
- Implemented advanced mathematical typesetting using KaTeX, supporting equation numbering, cross-referencing, and copy-to-clipboard functionality for complex technical articles.
- Developed type-safe API endpoints using Zod schema validation to ensure data integrity and robust error handling throughout the application backend.

Algorithmic Cryptocurrency Trading System
Jan 2020 – Dec 2020

Algorithmic crypto trading system integrating ML (forecasting), NLP (sentiment analysis), & quant finance across multiple exchanges. Features robust risk management, backtesting, and cloud-native design.

- Architected end-to-end trading infrastructure using CCXT/FreqTrade, implementing extensive backtesting and hyperparameter optimization frameworks validated across exchanges.
- Pioneered a hybrid ML forecasting approach combining time-series models (ARIMA) with deep learning (LSTMs, Transformers via TensorFlow/PyTorch), integrating technical, sentiment, and derivatives data.
- Engineered a sophisticated risk management system implementing dynamic position sizing via Sharpe ratio optimization and Markowitz portfolio theory for volatility-adjusted exposure.
- Developed a real-time market intelligence pipeline processing multi-source data (APIs, Reddit, Twitter, news) with NLP enhancements (OpenAI, LangChain, LlamaIndex) for sentiment-driven signals.

NBA Basketball Database
Apr 2021 – Jun 2021

High-performance data engineering solution processing 65k+ historical NBA games and 13M+ play-by-play events via cloud-native automated pipelines. Achieved 315k+ Kaggle dataset views and adoption in published research.

- Engineered a comprehensive NBA database using nba_api, ensuring data integrity across player, team, and game statistics with a robust Pandera schema validation system (15+ schemas).
- Designed automated daily/monthly update workflows using scheduled Kaggle notebooks and cron jobs, ensuring continuous data freshness and 99%+ completeness without dedicated infrastructure.
- Optimized SQLite database schema through normalization and foreign key relationships, enabling complex statistical analyses; maintained storage efficiency.
- Implemented advanced parallel processing using Python's multiprocessing Pool with dynamic scaling and intelligent retries, drastically reducing extraction time while managing API rate limits.

Regularized Linear Regression Deep Dive
Nov 2020 – Jan 2021

Implemented OLS, Ridge, Lasso, and Elastic Net regression algorithms from scratch in NumPy, including Pathwise Coordinate Descent optimization. Applied to wine quality prediction and published findings in a 3-part Towards Data Science series.

- Authored a high-impact 3-part series on Towards Data Science detailing the mathematical foundations and from-scratch implementations of regularized regression models.
- Implemented optimization algorithms like Pathwise Coordinate Descent from first principles, incorporating warm starts and covariance updates for performance comparable to standard libraries.
- Translated complex statistical theory (matrix calculus, optimization) into vectorized NumPy implementations for OLS, Ridge, Lasso, and Elastic Net, providing rigorous derivations.
- Engineered a complete analysis workflow including VIF calculation, K-Fold cross-validation, and visualization of regularization paths for model evaluation and interpretation.

NBA Game Attendance Forecaster
Oct 2019 – Jan 2020

Predictive analytics solution forecasting NBA game attendance with 95% accuracy (MAE ≈ 850 attendees). Utilized ensemble ML models (Random Forest, Gradient Boosting, AdaBoost) to enable data-driven venue operations and revenue optimization.

- Achieved 95% prediction accuracy (Mean Absolute Error ≈ 850 attendees, or 5% of venue capacity) using ensemble machine learning models including Random Forest, Gradient Boosting, and AdaBoost.
- Improved prediction accuracy by 20% through engineering and evaluating 3 distinct feature datasets combining NBA API data with external factors.
- Developed an automated ETL pipeline using the NBA API to ingest and process historical game data, team statistics, and attendance records.
- Ensured model reliability and generalizability using rigorous k-fold cross-validation methodology and statistical significance testing across performance metrics (R^2, MAE, RMSE).

SKILLS

Programming Languages:
Python · Java · SQL · JavaScript · TypeScript · R · Solidity · q (KDB+) · Bash/Shell · HTML5 · CSS3 · Matlab · Ruby

Frameworks & Libraries:
React · Next.js · Redux · Pandas · NumPy · Scikit-learn · PySpark · TensorFlow · PyTorch · FastAPI · Flask · Django REST · LangChain · LlamaIndex · Hugging Face Transformers · ethers.js · web3.js · SQLAlchemy · TailwindCSS · SCSS · shadcn/ui · Matplotlib · Seaborn · Plotly · Dash · Cypress · Jest · Pytest · Kedro · Dask · Ray · XGBoost · LightGBM · OR-Tools · CCXT · FreqTrade

Databases & Storage:
Oracle · KDB+ · PostgreSQL · MySQL · SQLite · Redis · MongoDB · Vector DBs (Weaviate) · AWS S3 · DynamoDB · Snowflake · BigQuery · Redshift · SQL · NoSQL · Timeseries DBs · Database Optimization · Schema Design · Data Modeling · Data Warehousing · Data Lakes · Parquet · Feather/Arrow · Delta Lake · Iceberg · Cassandra · Neo4j · ACID Transactions · Indexing

Cloud Platforms & Services:
AWS (EC2, S3, Lambda, SageMaker, EKS, Redshift, DynamoDB) · GCP (BigQuery, Cloud Functions, GKE) · Azure · Vercel · Serverless Architecture · Cloud-Native Development · Kubernetes · Docker · AWS Certified Cloud Practitioner · Infrastructure as Code (IaC) · Terraform · CloudFormation · CDK · Multi-region Deployment · High Availability · Scalability · CDN · API Gateway · Netlify · Heroku · Digital Ocean

Tools & Technologies:
Git · GitHub · Docker · Kubernetes · Apache Spark · Apache Kafka · Jira · Notion · CI/CD (GitHub Actions, Jenkins) · Linux · Shell Scripting · Jupyter · Conda · Poetry · Make · Swagger/OpenAPI · VS Code · Copilot · Codeium · LLMs (GPT, Claude) · MLflow · DVC · Grafana · Prometheus · Postman · Airflow · dbt · Prefect · Beam · Pandera · nba_api · Figma · Raspberry Pi · Arduino · Kaggle Notebooks

Methodologies & Concepts:
SDLC · OOP · Enterprise Architecture · Data Engineering (ETL/ELT) · Data Science · Machine Learning (ML) · Deep Learning (DL) · Natural Language Processing (NLP) · AI-Driven Development · Anomaly Detection · Time Series Forecasting · Agile (Scrum, Kanban) · TDD · REST API Design · Microservices · Web3/Blockchain (DeFi, NFTs, DAOs) · Operations Research (OR) · Mathematical Optimization (LP, IP, MILP) · Quantitative Finance · Algorithmic Trading · Risk Management · Portfolio Optimization · Data Visualization · Statistical Analysis · Performance Optimization · Web Rendering (SSR, SSG, ISR) · Observability · System Design · MLOps · Prompt Engineering (RAG) · Cloud-Native · Distributed Systems · Fair Division · Data Governance · Compliance · Technical Writing · Mentorship · Project Management

CERTIFICATES

- AWS Certified Cloud Practitioner — *Amazon Web Services (AWS)* Nov 2024
- IBM Data Science Professional Certificate — *IBM* Feb 2021

PUBLICATIONS

- Basics of Linear Regression Modeling and Ordinary Least Squares (OLS) — *Towards Data Science* Jan 2021
- Using Ridge Regression to Overcome Drawbacks of Ordinary Least Squares (OLS) — *Towards Data Science* Jan 2021
- Implementing Pairwise Coordinate Descent For The Lasso and The Elastic Net In Python — *Towards Data Science* Jan 2021